

25. Let A be an $n \times m$ matrix and let B be an $m \times n$ matrix. Prove that the matrix $I_m - BA$ is invertible if and only if the matrix $I_n - AB$ is invertible.

Solution: We shall prove that the matrix $I_m - BA$ is not invertible if and only if the matrix $I_n - AB$ is not invertible.

Let the matrix $I_m - BA$ be not invertible, so that $(I_m - BA)\mathbf{x} = \mathbf{0}$ has a non-trivial solution. Let $\mathbf{u} (\neq \mathbf{0})$ be such that $(I_m - BA)\mathbf{u} = \mathbf{0}$, that is, $\mathbf{u} = BA\mathbf{u}$. Now

$$\mathbf{u} = BA\mathbf{u} \Rightarrow A\mathbf{u} = ABA\mathbf{u} \Rightarrow (I_n - AB)A\mathbf{u} = \mathbf{0}.$$

Moreover, $A\mathbf{u} \neq \mathbf{0}$ since $\mathbf{u} = BA\mathbf{u}$ and $\mathbf{u} \neq \mathbf{0}$. Thus $(I_n - AB)\mathbf{x} = \mathbf{0}$ has a non-trivial solution, and hence the matrix $I_n - AB$ is not invertible.

Similarly, if the matrix $I_n - AB$ is not invertible then the matrix $I_m - BA$ is also not invertible.

Aliter

Suppose that $I_n - AB$ is invertible. Let $C = B(I_n - AB)^{-1}A + I_m$. Then

$$\begin{aligned} (I_m - BA)C &= I_m - BA + B(I_n - AB)^{-1}A - BAB(I_n - AB)^{-1}A \\ &= I_m - BA + (B - BAB)(I_n - AB)^{-1}A \\ &= I_m - BA + B(I_n - AB)(I_n - AB)^{-1}A = I_m - BA + BA = I_m. \end{aligned}$$

Hence $I_m - BA$ is invertible. Similarly, if the matrix $I_m - BA$ is invertible then the matrix $A(I_m - BA)^{-1}B + I_n$ is the inverse of $I_n - AB$. $\square$

26. Let A be a nilpotent matrix (*i.e.*, $A^m = \mathbf{0}$ for some $m \geq 1$). Show that the matrix $I + A$ is invertible.

Solution: Take $B = I - A + A^2 - A^3 + \dots + (-1)^{m-1}A^{m-1}$. Then we see that

$$\begin{aligned} (I + A)B &= (I + A)[I - A + A^2 - A^3 + \dots + (-1)^{m-1}A^{m-1}] \\ &= [I - A + A^2 - A^3 + \dots + (-1)^{m-1}A^{m-1}] + \\ &\quad [A - A^2 + A^3 - A^4 + \dots + (-1)^{m-2}A^{m-1}], \text{ since } A^m = \mathbf{0} \\ &= I. \end{aligned}$$

Hence $I + A$ is invertible, and $B = I - A + A^2 - A^3 + \dots + (-1)^{m-1}A^{m-1}$ is the inverse of $I + A$.

Aliter

Consider the system $(I + A)\mathbf{x} = \mathbf{0}$, *i.e.*, $\mathbf{x} + A\mathbf{x} = \mathbf{0}$. Then we get $\mathbf{x} = -A\mathbf{x}$, and so

$$\mathbf{x} = -A\mathbf{x}, A\mathbf{x} = -A^2\mathbf{x}, A^2\mathbf{x} = -A^3\mathbf{x}, \dots, A^{m-2}\mathbf{x} = -A^{m-1}\mathbf{x}, A^{m-1}\mathbf{x} = -A^m\mathbf{x} = \mathbf{0}, \text{ as } A^m = \mathbf{0}.$$

This gives that $\mathbf{x} = \mathbf{0}$. Thus the system $(I + A)\mathbf{x} = \mathbf{0}$ has only the trivial solution, and hence the matrix $I + A$ is invertible. $\square$

27. Let A and B be two $n \times n$ matrices. If AB is invertible then show that each of the matrices A and B must be invertible, in at least three different ways.

Solution: **Method 1:** Use $\det(AB) = \det(A)\det(B)$. [Based on Section 5]

Method 2: Use $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$. [Based on Section 6]

Method 3: Let AB be invertible and let C be the inverse of AB . Then we have $(AB)C = I \Rightarrow A(BC) = I \Rightarrow A$ is invertible. Again, $C(AB) = I \Rightarrow (CA)B = I \Rightarrow B$ is invertible.

Method 4: Let $\mathbf{x} \in \mathbb{R}^n$. Then we have,

$$B\mathbf{x} = \mathbf{0} \Rightarrow AB\mathbf{x} = A\mathbf{0} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0}, \text{ since } AB \text{ is invertible.}$$

Thus the system $B\mathbf{x} = \mathbf{0}$ has only the trivial solution, and hence the matrix B is invertible. Finally, since product of two invertible matrices is invertible, $A = (AB)B^{-1}$ is also invertible. $\square$

28. Let A be an $n \times n$ skew-symmetric matrix. Prove that $\mathbf{x}^t A \mathbf{x} = 0$ for all $\mathbf{x} \in \mathbb{R}^n$. Further, prove that the matrix $I_n + A$ is invertible.

Solution: Let $A = [a_{ij}]$ and $\mathbf{x} = [x_1, x_2, \dots, x_n]^t \in \mathbb{R}^n$. Then we have

$$\mathbf{x}^t A \mathbf{x} = \sum_{i=1}^n a_{ii} x_i^2 + \sum_{1 \leq i < j \leq n} (a_{ij} + a_{ji}) x_i x_j.$$

If A is an skew-symmetric matrix then we have $a_{ii} = 0$ and $a_{ij} = -a_{ji}$. Hence $\mathbf{x}^t A \mathbf{x} = 0$.

OR As $\mathbf{x}^t A \mathbf{x}$ is a number and $A^t = -A$, we have

$$\mathbf{x}^t A \mathbf{x} = (\mathbf{x}^t A \mathbf{x})^t = \mathbf{x}^t A^t \mathbf{x} = -\mathbf{x}^t A \mathbf{x} \Rightarrow 2\mathbf{x}^t A \mathbf{x} = 0 \Rightarrow \mathbf{x}^t A \mathbf{x} = 0.$$

Let $\mathbf{x} \in \mathbb{R}^n$. Then we have

$$(I_n + A)\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x}^t(I_n + A)\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x}^t \mathbf{x} + \mathbf{x}^t A \mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x}^t \mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0}.$$

Thus the system $(I_n + A)\mathbf{x} = \mathbf{0}$ has only the trivial solution, and hence the matrix $I_n + A$ is invertible. $\square$

29. Check, without calculating the actual value, whether the determinant of the following matrix is even or odd:

$$\begin{bmatrix} 2 & 2 & 5 & 4 \\ 5 & 3 & 2 & 2 \\ 2 & 5 & 7 & 5 \\ 1 & 3 & 9 & 2 \end{bmatrix}.$$

Solution: An integer is even or odd according as the remainder, after division by 2, is 0 or 1. Replacing all the even numbers by 0 and the odd numbers by 1, the given matrix reduces to

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}.$$

Adding the first two rows to the fourth row, and then replacing 2 by 0, we get

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

whose determinant is zero. Hence the determinant of the given matrix is **even**.

Aliter:

Adding the first two rows to the fourth row in the given matrix, we get

$$\begin{vmatrix} 2 & 2 & 5 & 4 \\ 5 & 3 & 2 & 2 \\ 2 & 5 & 7 & 5 \\ 8 & 8 & 16 & 8 \end{vmatrix} = 8 \cdot \begin{vmatrix} 2 & 2 & 5 & 4 \\ 5 & 3 & 2 & 2 \\ 2 & 5 & 7 & 5 \\ 1 & 1 & 2 & 1 \end{vmatrix}.$$

Hence the determinant of the given matrix is **even**. □

30. Find the determinant of the following matrices:

$$\begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \dots & x_2^{n-1} \\ \dots & \dots & \dots & \dots & \dots \\ 1 & x_n & x_n^2 & \dots & x_n^{n-1} \end{bmatrix} \text{ and } \begin{bmatrix} 1 & 2 & 3 & 4 & \dots & n \\ 2 & 2 & 3 & 4 & \dots & n \\ 3 & 3 & 3 & 4 & \dots & n \\ \dots & \dots & \dots & \dots & \dots & \dots \\ n & n & n & n & \dots & n \end{bmatrix}.$$

Solution: For $n \geq 2$, let A_n denote the first given matrix (this matrix is known as **Vandermonde** Matrix). We shall use method of induction on n to show $\det(A_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i)$.

We can assume that all x_i 's are distinct, otherwise $\det(A_n) = 0$.

For $n = 2$, clearly $\det(A_2) = x_2 - x_1 = \prod_{1 \leq i < j \leq 2} (x_j - x_i)$. Assume $\det(A_{k-1}) = \prod_{1 \leq i < j \leq (k-1)} (x_j - x_i)$,

where $k \geq 3$. Now we have

$$\begin{aligned} \det(A_k) &= \begin{vmatrix} 1 & x_1 & x_1^2 & \dots & x_1^{k-1} \\ 1 & x_2 & x_2^2 & \dots & x_2^{k-1} \\ \dots & \dots & \dots & \dots & \dots \\ 1 & x_k & x_k^2 & \dots & x_k^{k-1} \end{vmatrix} \\ &= \begin{vmatrix} 0 & x_1 - x_k & x_1^2 - x_k^2 & \dots & x_1^{k-1} - x_k^{k-1} \\ 0 & x_2 - x_k & x_2^2 - x_k^2 & \dots & x_2^{k-1} - x_k^{k-1} \\ \dots & \dots & \dots & \dots & \dots \\ 0 & x_{k-1} - x_k & x_{k-1}^2 - x_k^2 & \dots & x_{k-1}^{k-1} - x_k^{k-1} \\ 1 & x_k & x_k^2 & \dots & x_k^{k-1} \end{vmatrix} \quad (\text{applying } R_i \rightarrow R_i - R_k \text{ for each } i < k) \\ &= (-1)^{k+1} \begin{vmatrix} x_1 - x_k & x_1^2 - x_k^2 & \dots & x_1^{k-1} - x_k^{k-1} \\ \dots & \dots & \dots & \dots \\ x_{k-1} - x_k & x_{k-1}^2 - x_k^2 & \dots & x_{k-1}^{k-1} - x_k^{k-1} \end{vmatrix} \quad (\text{expanding through the 1st column}) \end{aligned}$$

$$\begin{aligned}
&= (-1)^{k+1} (x_1 - x_k) \dots (x_{k-1} - x_k) \begin{vmatrix} 1 & x_1 + x_k & \dots & x_1^{k-2} + x_1^{k-3}x_k + \dots + x_k^{k-2} \\ \dots & \dots & \dots & \dots \\ 1 & x_{k-1} + x_k & \dots & x_{k-1}^{k-2} + x_{k-1}^{k-3}x_k + \dots + x_k^{k-2} \end{vmatrix} \\
&= (x_k - x_1) \dots (x_k - x_{k-1}) \begin{vmatrix} 1 & x_1 & \dots & x_1^{k-2} \\ \dots & \dots & \dots & \dots \\ 1 & x_{k-1} & \dots & x_{k-1}^{k-2} \end{vmatrix} \quad (\text{applying } C_i \rightarrow C_i - x_k C_{i-1}, \text{ for each } i \geq 2) \\
&\quad \text{first for } i = k-1, \text{ then for } i = k-2, \text{ and so on} \\
&= (x_k - x_1) \dots (x_k - x_{k-1}) \prod_{1 \leq i < j \leq k-1} (x_j - x_i) \quad (\text{by induction hypothesis}) \\
&= \prod_{1 \leq i < j \leq k} (x_j - x_i).
\end{aligned}$$

Hence, by induction we conclude that $\det(A_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i)$.

Let the second given matrix be B . Applying the sequence $R_n \rightarrow R_n - R_{n-1}, R_{n-1} \rightarrow R_{n-1} - R_{n-2}, \dots, R_2 \rightarrow R_2 - R_1$ of elementary row operations on B , the matrix B is transformed to

$$\begin{bmatrix} 1 & 2 & 3 & 4 & \dots & n \\ 1 & 0 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 1 & 1 & 1 & 1 & \dots & 0 \end{bmatrix}.$$

Let B_{1n} be the matrix obtained by deleting the first row and the last column of the above matrix. Then B_{1n} will be a lower triangular matrix with all diagonal entries equal to 1.

Hence $\det(B) = (-1)^{1+n} n \cdot \det(B_{1n}) = (-1)^{n+1} n$. $\square$

31. Let $A = [a_{ij}]$ be an $n \times n$ matrix such that $\sum_{j=1}^n a_{ij} = 1$ for all $i = 1, 2, \dots, n$. Is it true that $\det(A - I_n) = 0$? Justify your answer.

Solution: As $\sum_{j=1}^n a_{ij} = 1$ for all $i = 1, 2, \dots, n$, the sum of the entries of each row of $A - I_n$ will be zero. Therefore, the column operation $C_1 \rightarrow \sum_{i=1}^n C_i$ makes the first column of $A - I_n$ zero, and hence $\det(A - I_n) = 0$. $\square$

32. Let $A = [a_{ij}]$ be an invertible matrix and let $B = [k^{i-j} a_{ij}]$, where $k (\neq 0) \in \mathbb{R}$. Find B^{-1} and $\det(B)$.

Solution: We notice that

$$B = \begin{bmatrix} k & 0 & \dots & 0 \\ 0 & k^2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & k^n \end{bmatrix} A \begin{bmatrix} k^{-1} & 0 & \dots & 0 \\ 0 & k^{-2} & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & k^{-n} \end{bmatrix}.$$

Hence

$$\begin{aligned}
B^{-1} &= \begin{bmatrix} k^{-1} & 0 & \dots & 0 \\ 0 & k^{-2} & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & k^{-n} \end{bmatrix}^{-1} A^{-1} \begin{bmatrix} k & 0 & \dots & 0 \\ 0 & k^2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & k^n \end{bmatrix}^{-1} \\
&= \begin{bmatrix} k & 0 & \dots & 0 \\ 0 & k^2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & k^n \end{bmatrix} A^{-1} \begin{bmatrix} k^{-1} & 0 & \dots & 0 \\ 0 & k^{-2} & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & k^{-n} \end{bmatrix}.
\end{aligned}$$

Thus, if $A^{-1} = [x_{ij}]$ then $B^{-1} = [k^{i-j}x_{ij}]$.

We have

$$\det(B) = \begin{vmatrix} k & 0 & \dots & 0 \\ 0 & k^2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & k^n \end{vmatrix} \det(A) \begin{vmatrix} k^{-1} & 0 & \dots & 0 \\ 0 & k^{-2} & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & k^{-n} \end{vmatrix} = k^{\frac{n(n+1)}{2}} \cdot \det(A) \cdot k^{-\frac{n(n+1)}{2}} = \det(A).$$

□